

Folding and Interpolating ADC

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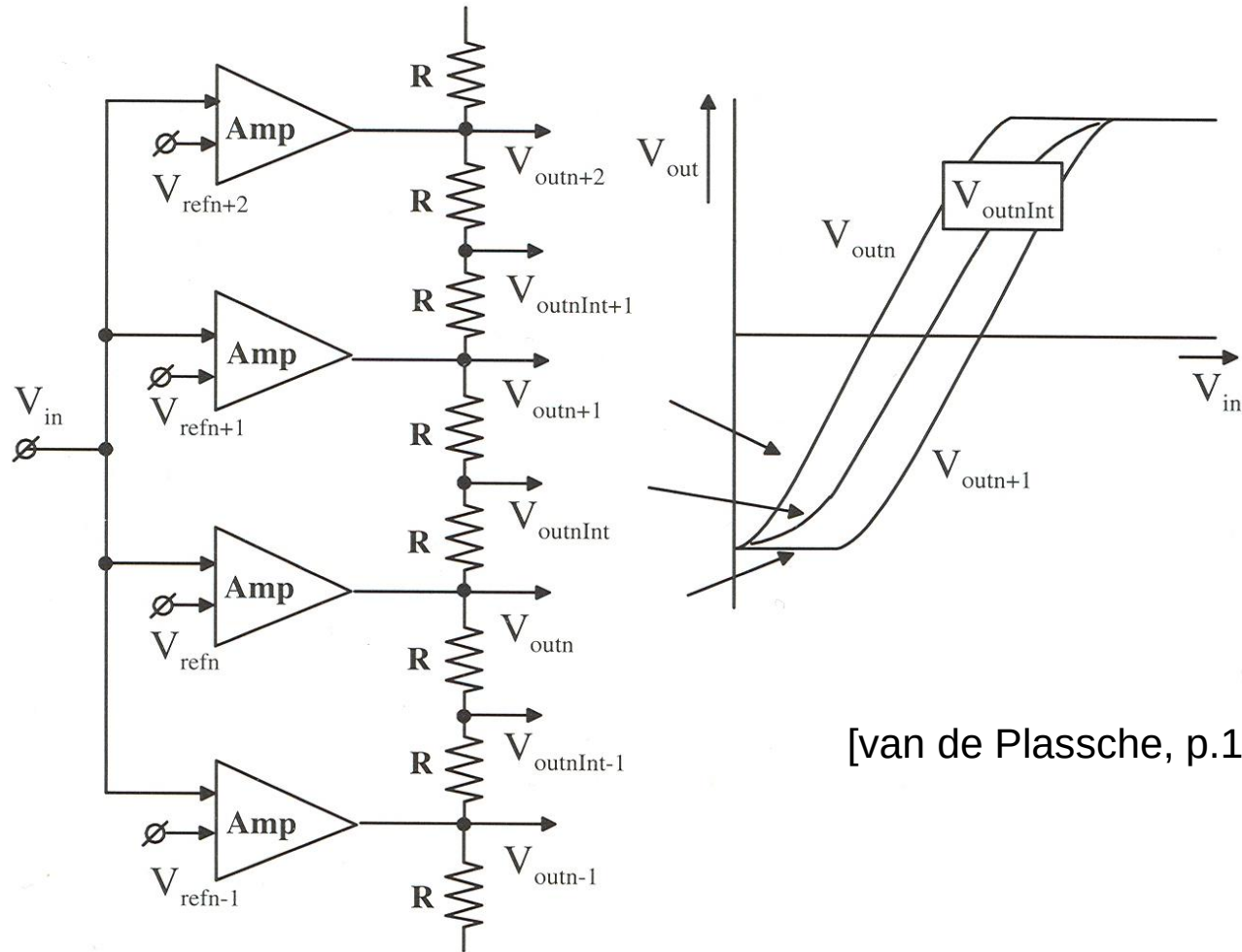
Reducing Complexity

- How much power would a 8-bit flash ADC consume?
 - A 6-bit 1GS/s flash ADC consumes about 500mW (in 0.35um)
 - To increase the resolution to 8-bit, we need to
 - Increase the number of comparators by 4
 - Increase size & power of each comparator by 4^2 (matching)
 - **ADC Power: $500\text{mW} \cdot 4^3 = 32\text{W}$ (!)**
- How to increase resolution w/o significantly increasing power?
 - Interpolation
 - Folding
 - Folding & interpolation
 - Multi-step conversion, pipelining

Interpolation

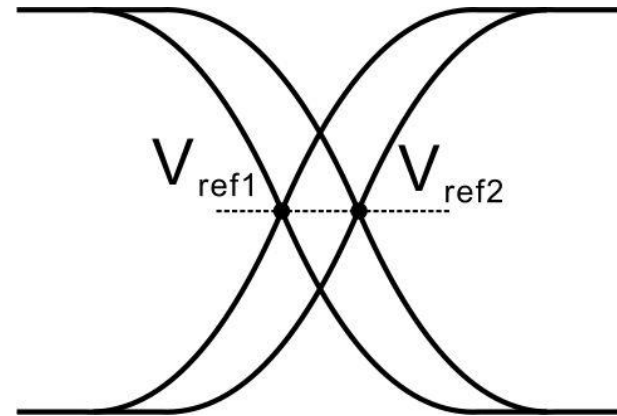
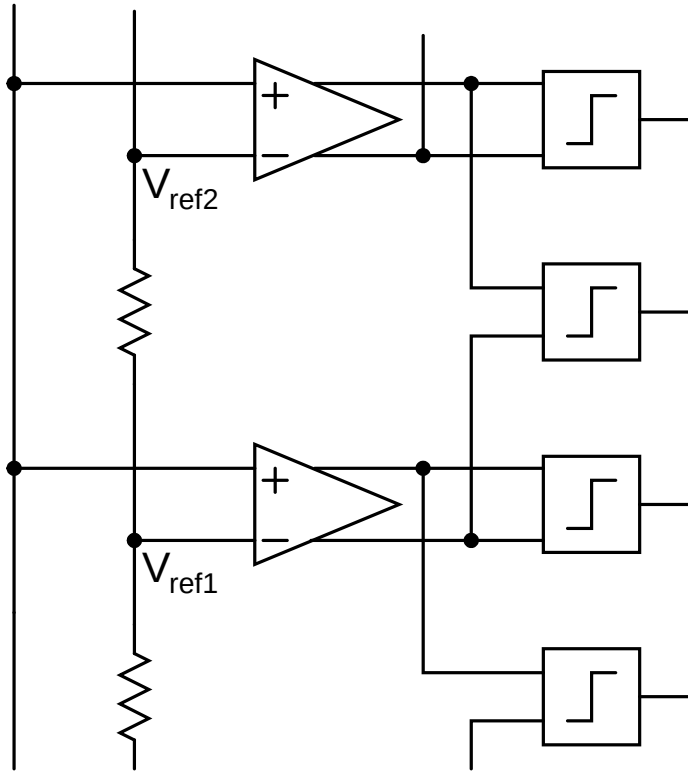
- Reduce # preamps by interpolating preamp outputs
 - Lower power dissipation
 - Lower input capacitance
 - Lower area
- Same number of latches
- Important “side-benefit”
 - Lower offset through offset averaging

Concept

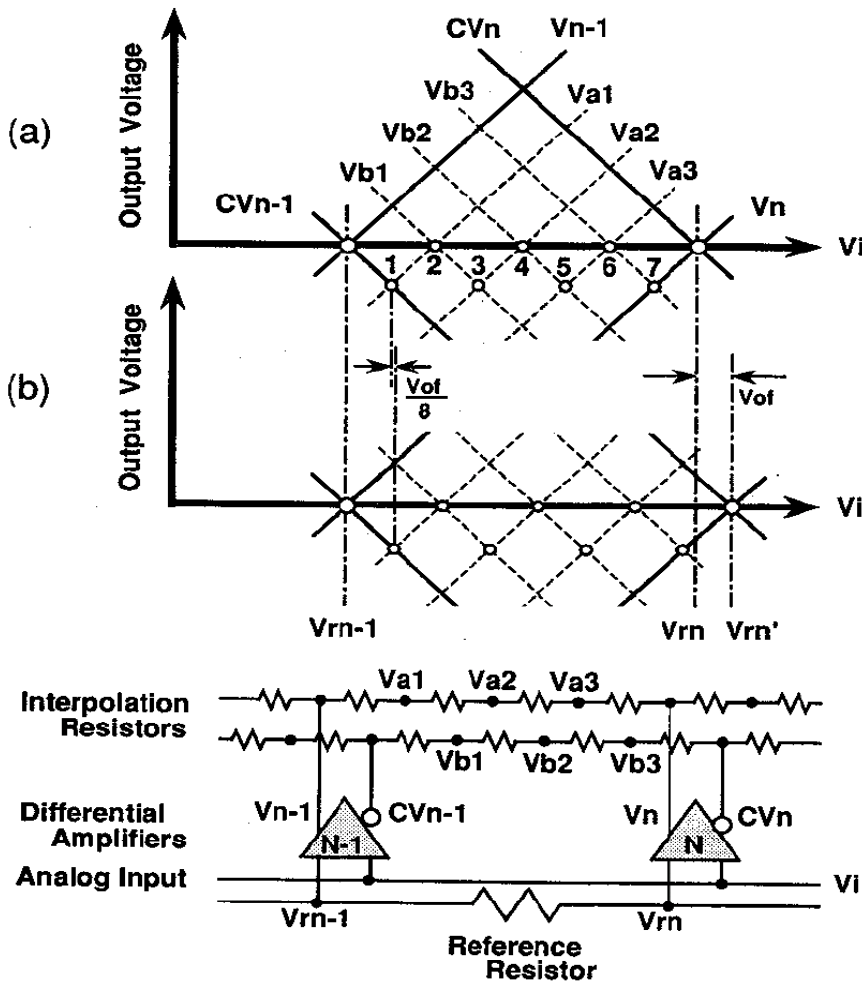


[van de Plassche, p.118]

Differential Implementation



Higher Order Interpolation



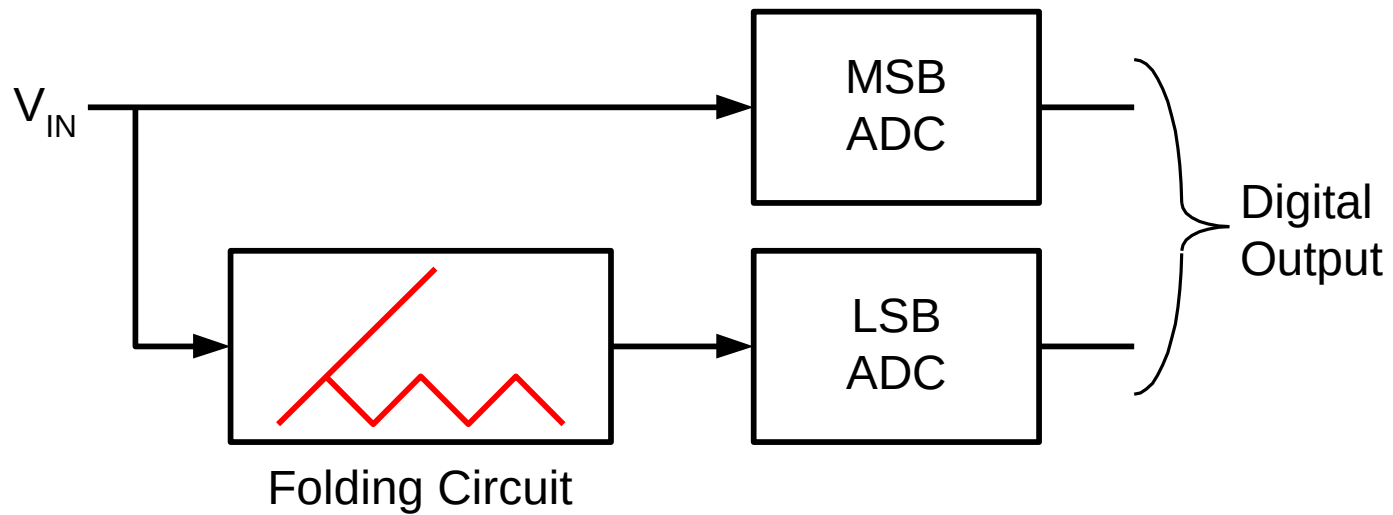
- Resistors produce additional levels
- Factor = # latch / # preamp
- Factor = 8 for this example

H. Kimura et al, "A 10-b 300-MHz Interpolated-Parallel A/D Converter," IEEE J. of Solid-State Circuits, pp. 438-446, April 1993.

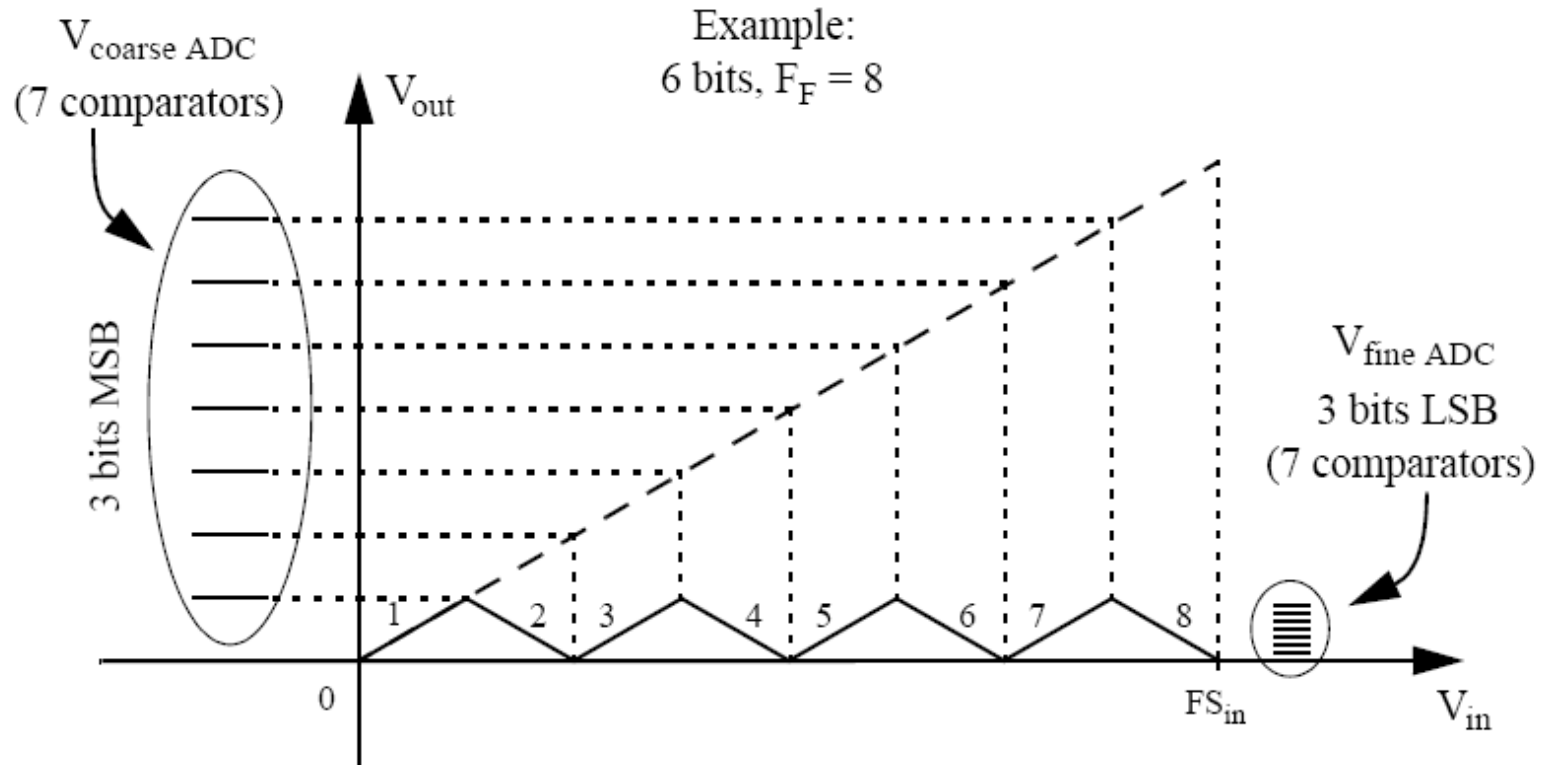
Issues with Interpolation

- Speed penalty
 - Resistor ladder resistance
 - Latch capacitance and parasitic capacitance
- Preamplifier nonlinearity
 - For interpolation factors >2 , amplifier nonlinearity can limit the precision of zero crossings
 - Must ensure that "linear range" of adjacent preamplifiers overlaps
- Upper bound on interpolation factor, typically around 4
- Interpolation factor of 2 is widely used
 - No need for resistor string, low design complexity
- Details see [van de Plassche, p.121]

Folding ADC

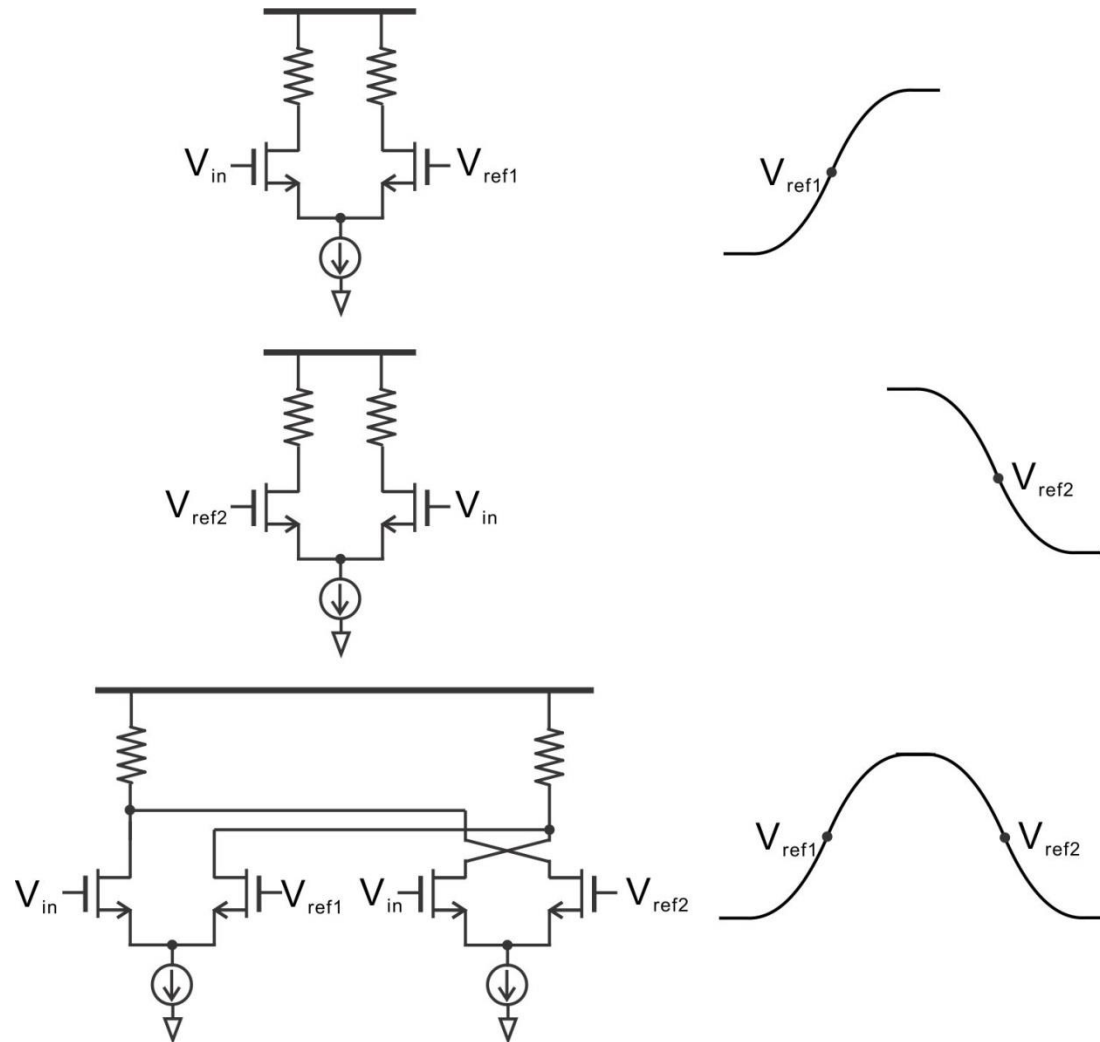


Example: 6-bit Folding ADC

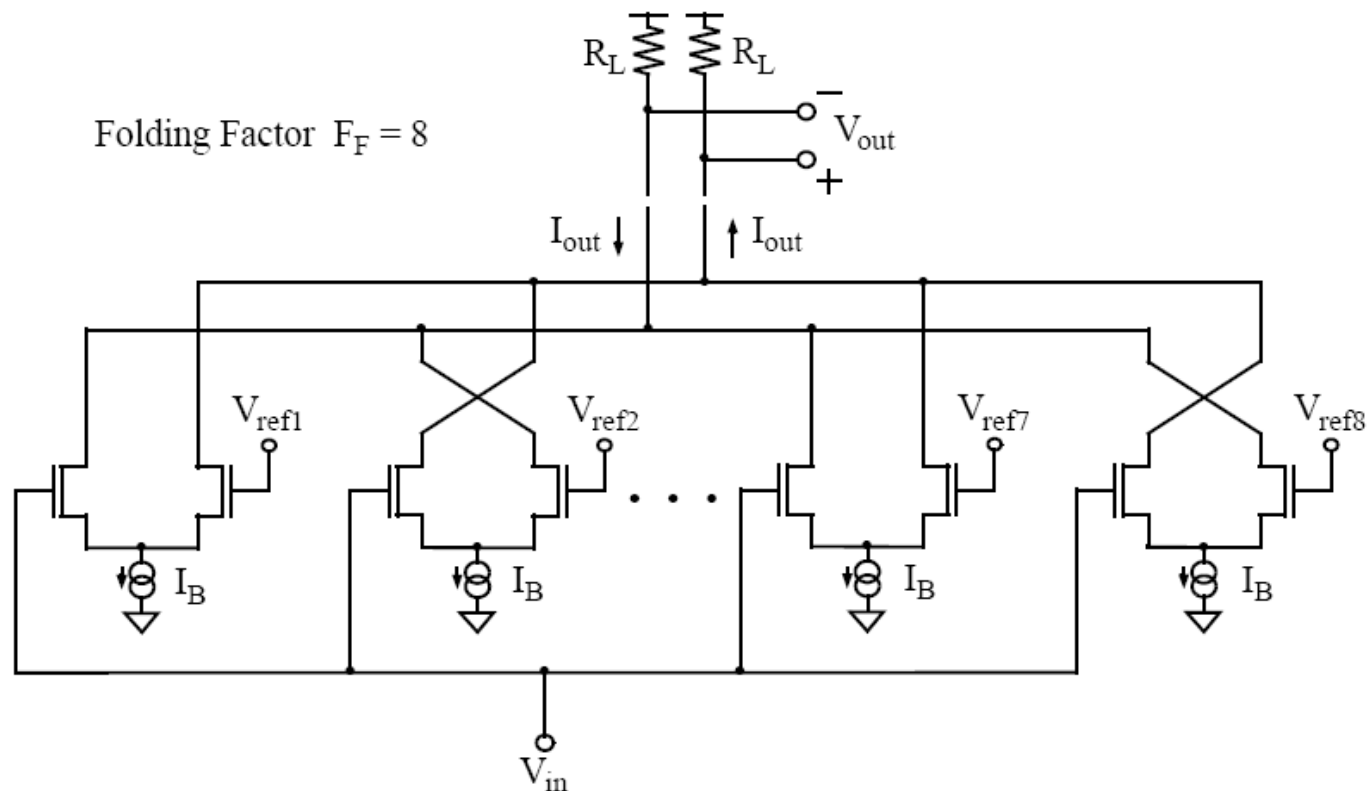


- Coarse ADC determines segment, fine ADC determines level within segment
- Folding factor (F_F) quantifies number of segments

Folder



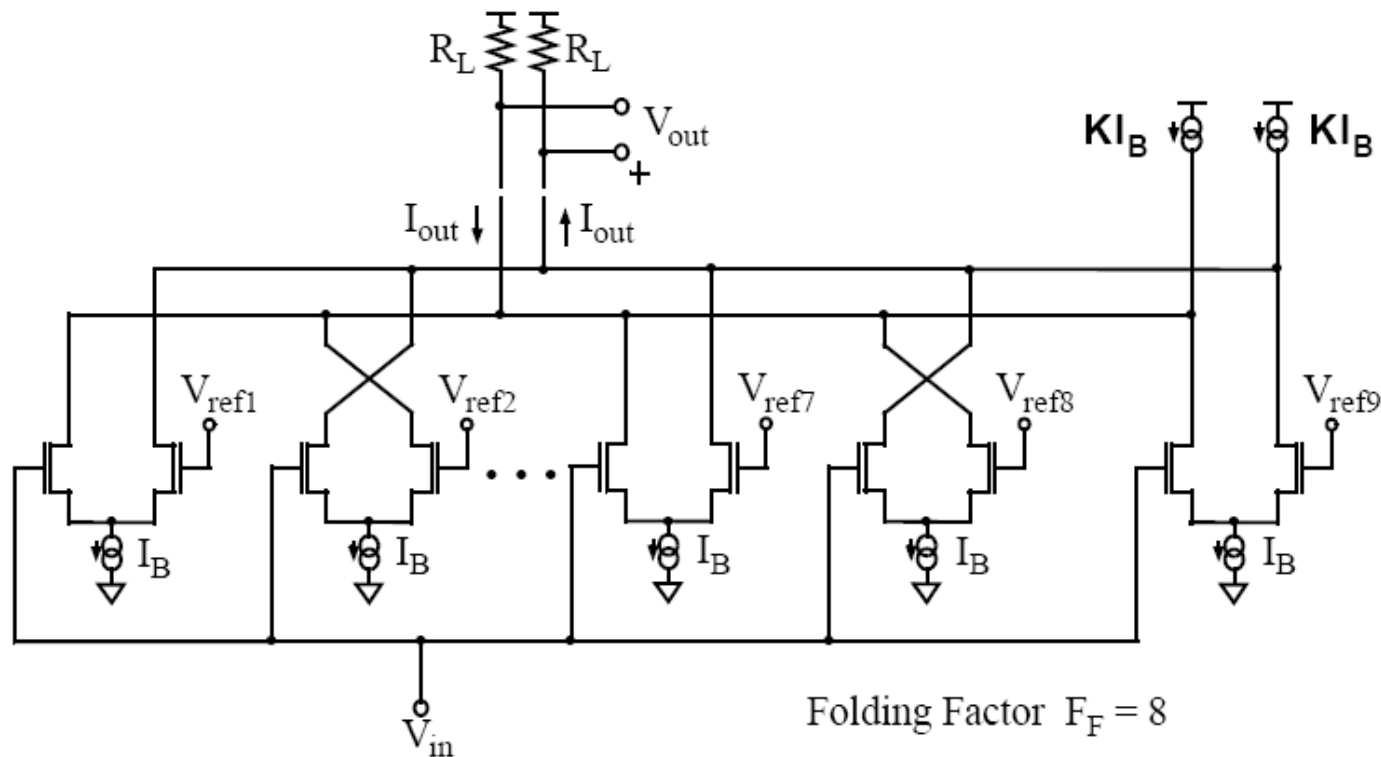
Folder Realization



- For any V_{in} , only one differential pair is "active", all others are saturated
- $V_{out, cm} = V_{dd} - F_F \times I_B R_L / 2$

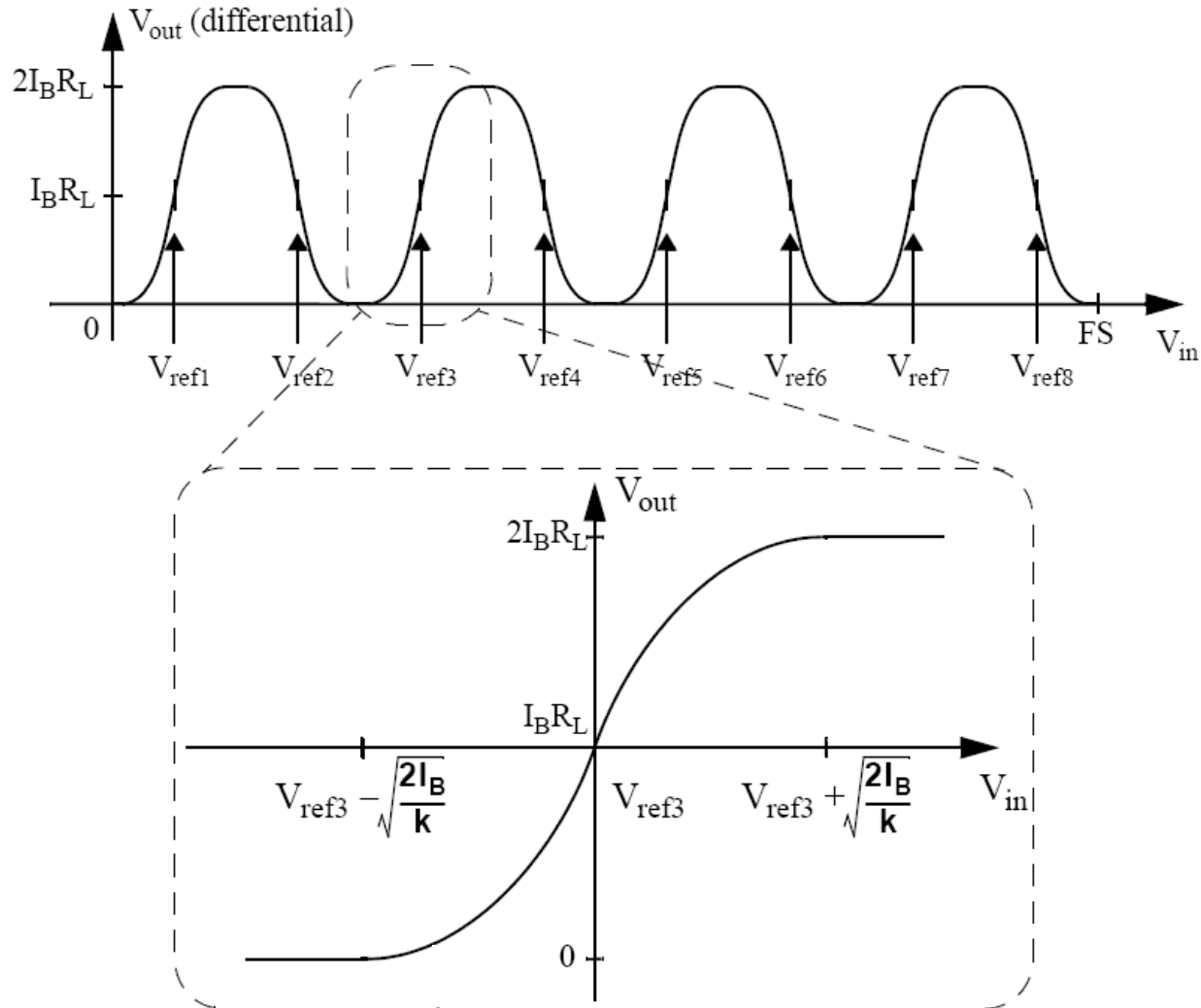
- $V_{\text{out, cm}} = V_{\text{dd}} - F_F \times I_B R_L / 2$

Improved Realization

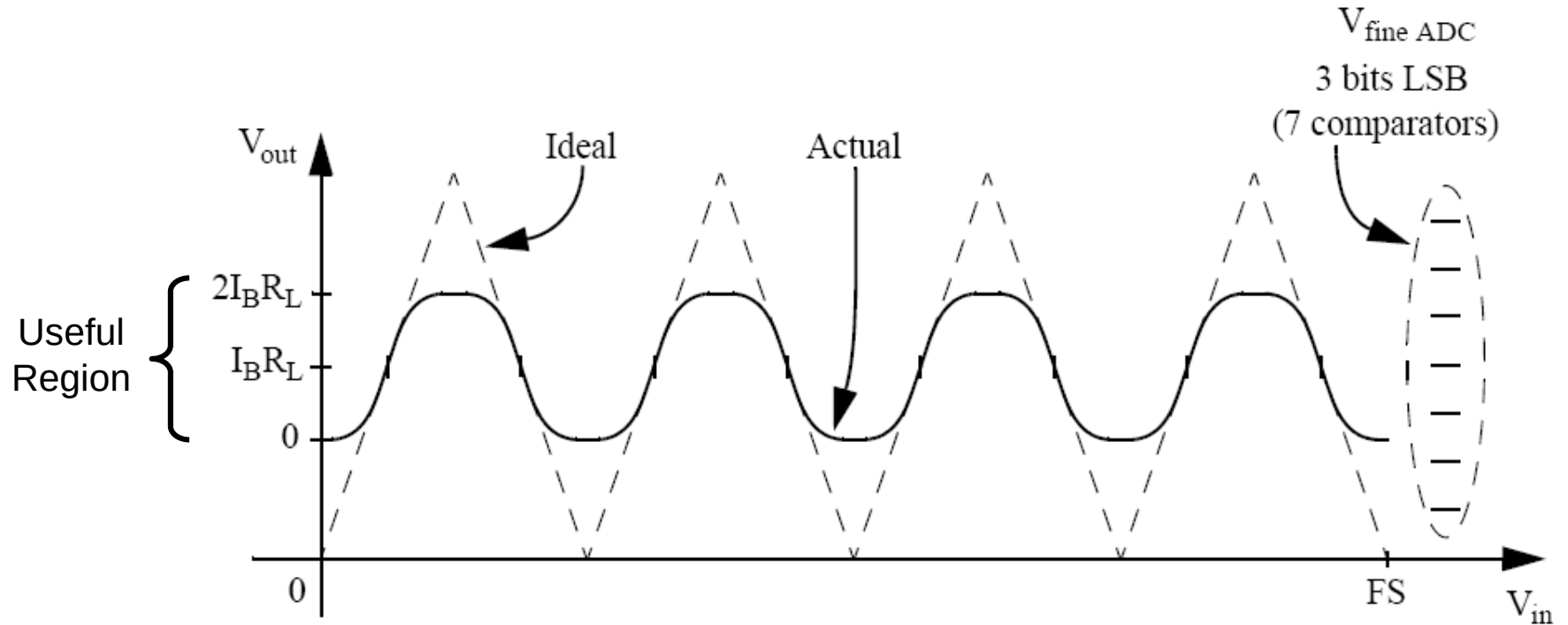


- Extra differential pair controls $V_{out, cm}$
- $V_{out, cm} = V_{dd} - (F_F + 1) \times I_B R_L / 2 + K \times I_B R_L$

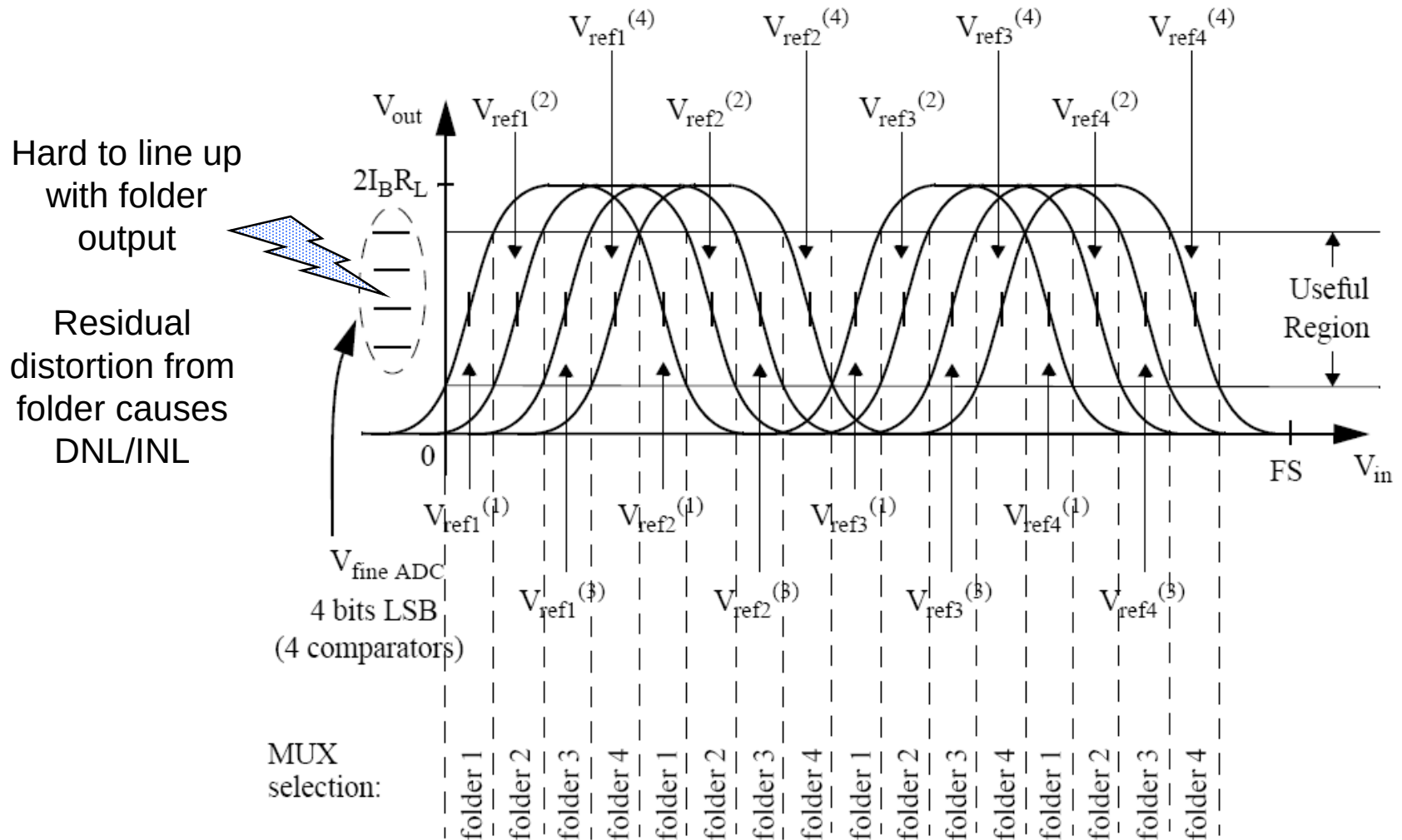
Input-Output Characteristic



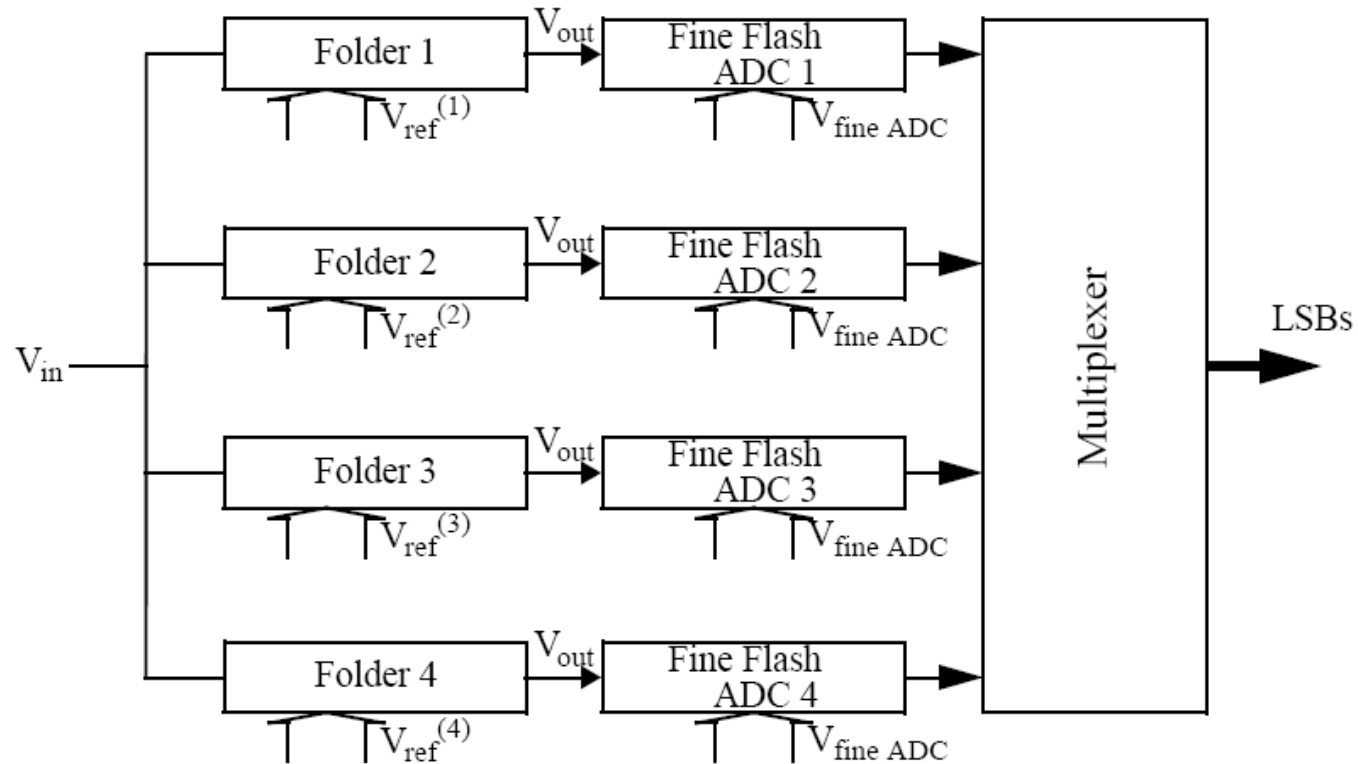
Rounding Problem



Multiple Folds (1)

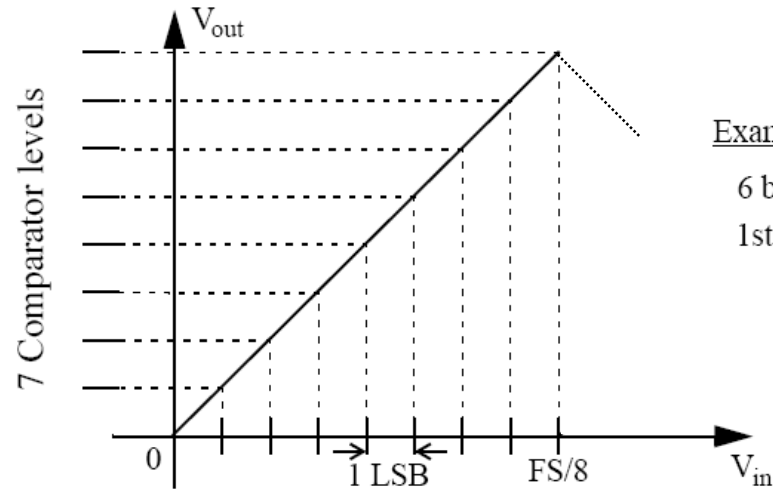


Multiple Folds (2)



- Use several folders so that any input value falls into useable "linear" region

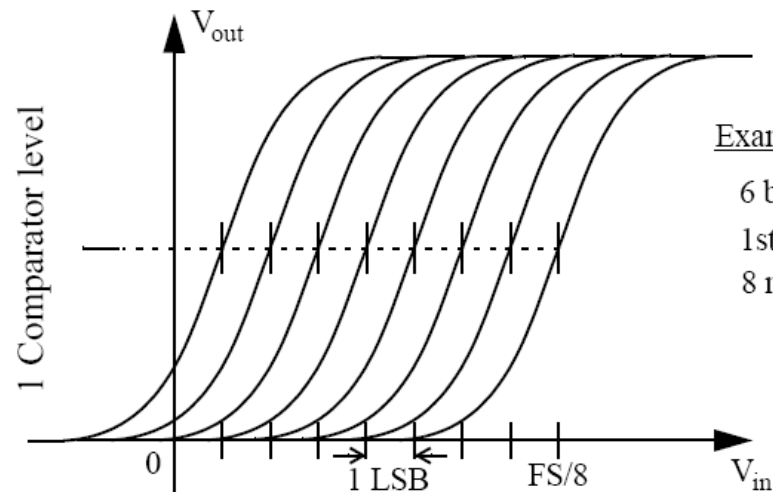
Multiple Folds Using Single Threshold (1)



Example 1:

6 bits, $F_F = 8$

1st segment of an ideal fold

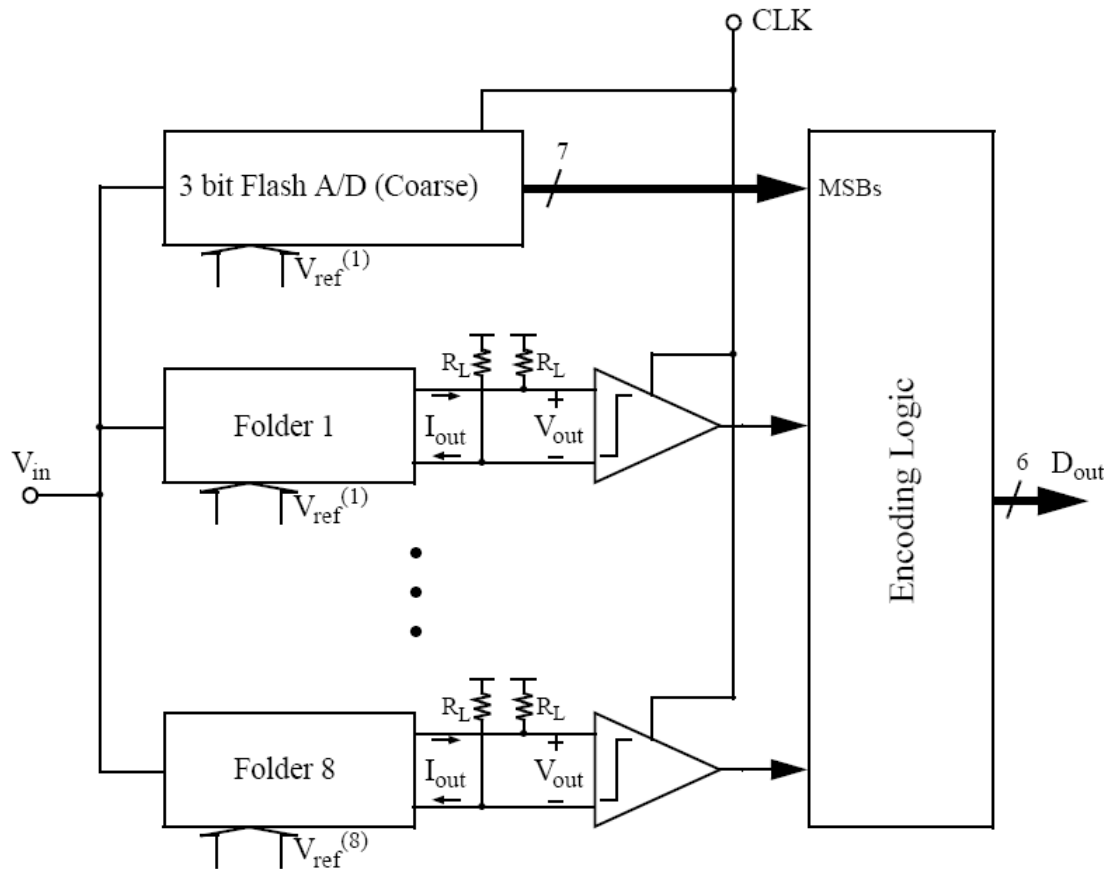


Example 2:

6 bits, $F_F = 8$

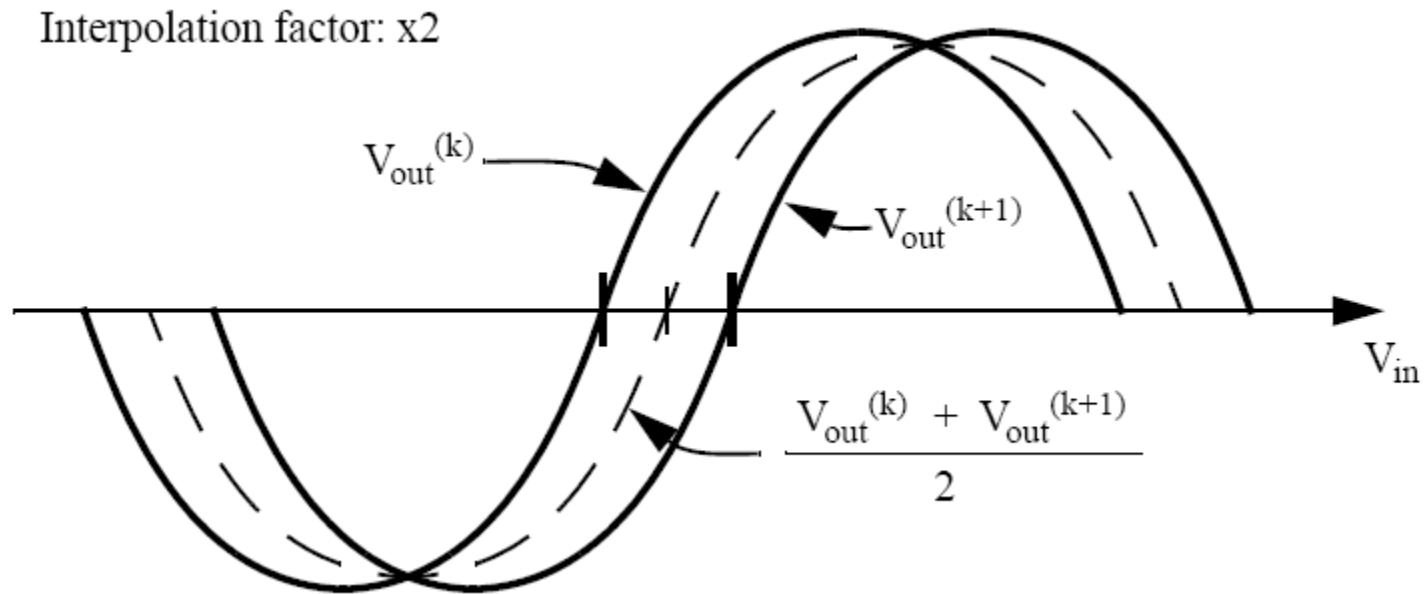
1st segment of
8 non-ideal folds

Multiple Folds Using Single Threshold (2)



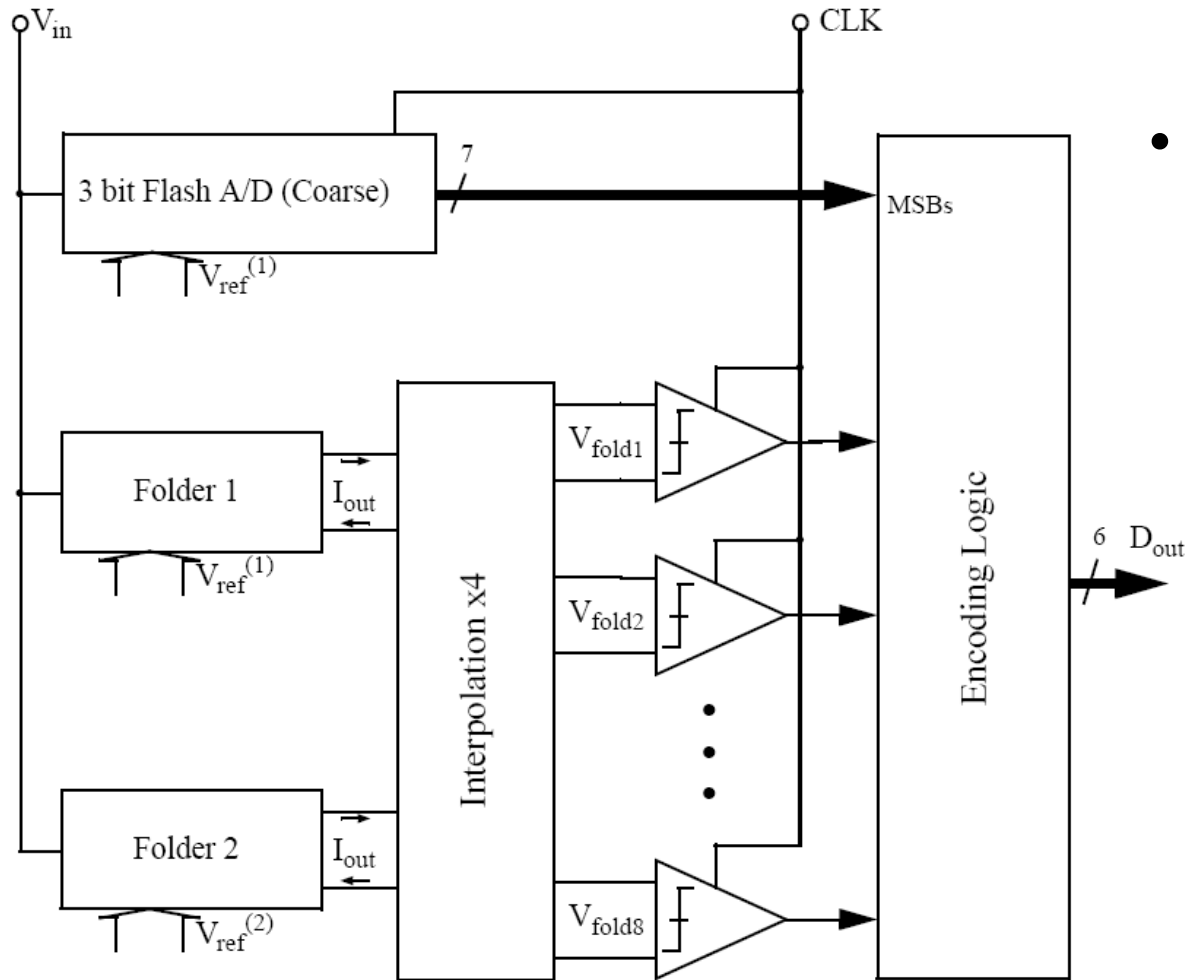
- Initial plan
 - Use one folder and 7 comparators in LSB section
- Now have
 - 8 folders and 8 comparators
- Reduce complexity
 - Use interpolation to eliminate some of the folders

Interpolation



- Same idea as discussed in the context of flash ADCs

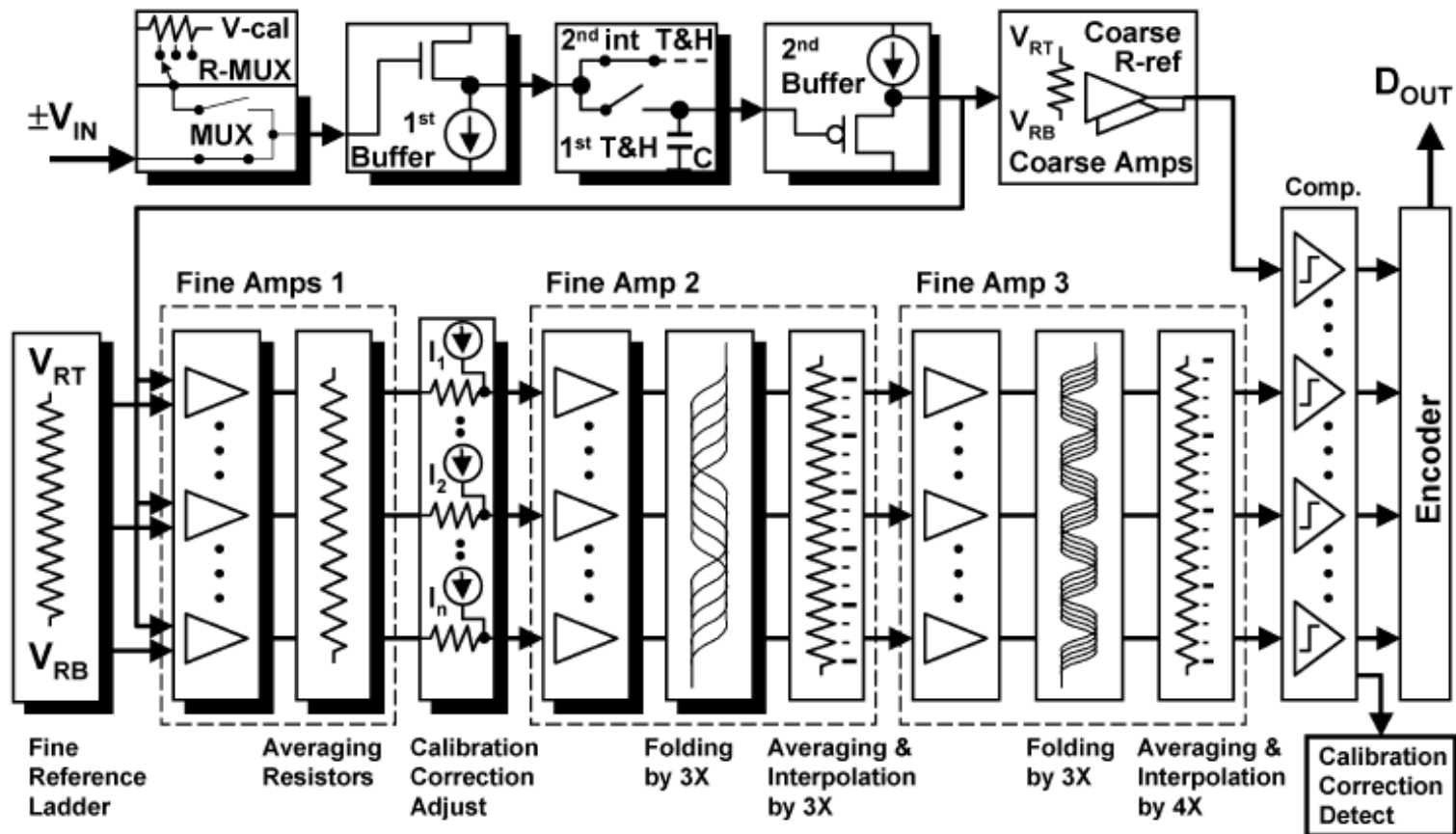
Complete Folding & Interpolating ADC



- 6-bit Example
 - 2 folders
 - 4x interpolation
 - $F_F = 8$

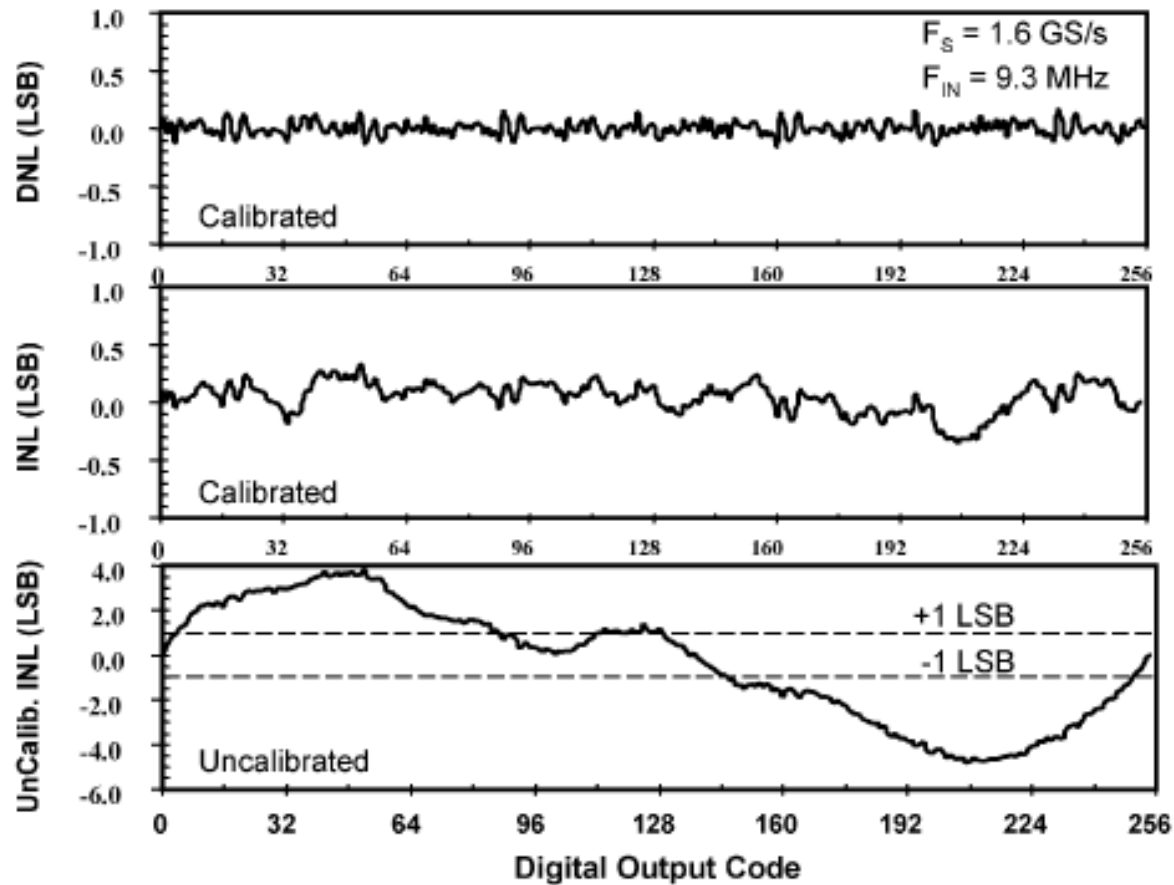
State-of-the-art Implementation (1)

[Taft et al., JSSC 12/2004]

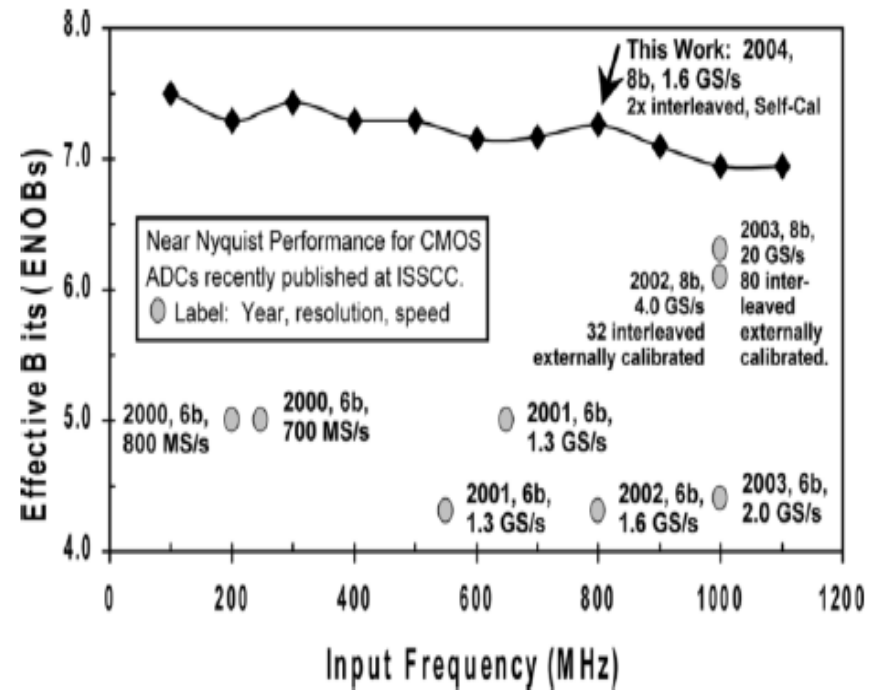
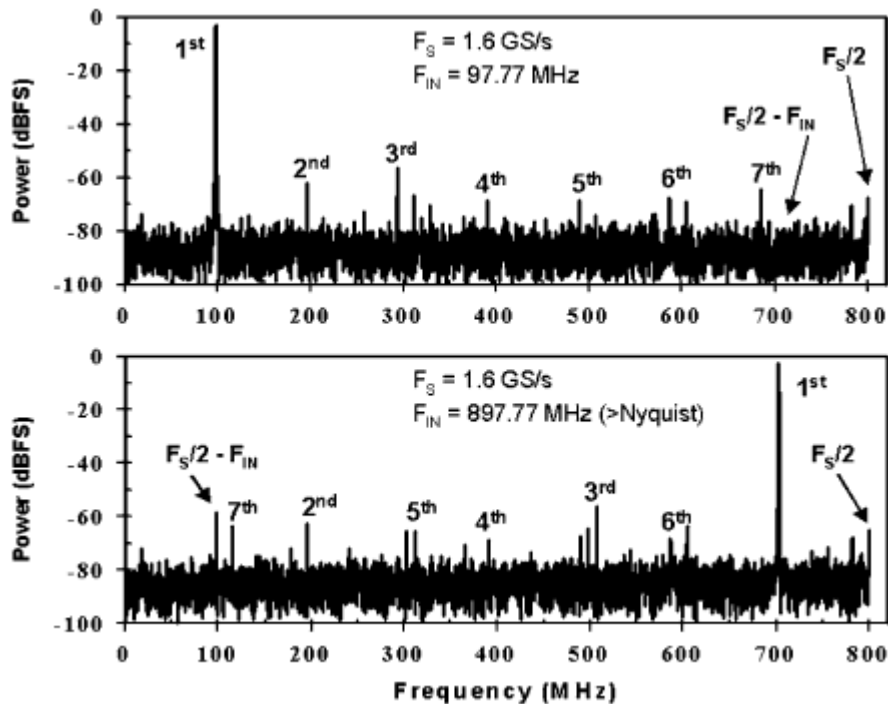


- Use calibration
- Use track and hold (let it settle)

State-of-the-art Implementation (2)



State-of-the-art Implementation (3)



State-of-the-art Implementation (4)

	$F_{IN} = 97.77 \text{ MHz}$	$F_{IN} = 797.77 \text{ MHz (Nyquist)}$
Sample Rate, F_S	1.6 GS/s	
Resolution	8 bits	
Max DNL	$\pm 0.15 \text{ LSB}$	
Max INL	$\pm 0.35 \text{ LSB}$	
SNR	48 dB	46 dB
SFDR	61 dB	56 dB
THD	- 57 dB	- 57 dB
ENOBs	7.60	7.26
Interleave aperture offset	$< 0.35 \text{ ps @ } F_S = 1 \text{ GS/s \& } F_{IN} = 1.5 \text{ GHz}$	
Input (-3 dB) Bandwidth	$> 1.75 \text{ GHz}$	
Resolution (-0.5 ENOB) Bandwidth	1.0 GHz	
Input Range	$\pm 400 \text{ mV differential}$	
Input Capacitance	1.8 pF (to gnd, w/o package)	
Input Termination	50 Ω (100 Ω differential)	
Single Supply	1.8 V	
Analog (DC) Current	245 mA	
Switching (AC) Current	185 mA	
LVDS Output Drivers	90 mA	
ADC Core Power (w/o outputs)	774 mW	
ADC core area	3.6 mm ²	
ADC die area	16 mm ² (for dual ADC, pad limited)	
Package	128-pin EPQFP	
Technology	0.18 μm CMOS (1-poly, 5-metal) No capacitor module nor dual-gate process	

Folding ADC Problems & Solutions

- Dynamic problems
 - Frequency at the folder output is approximately input frequency times folding factor!
 - Hard to match the delay through coarse/fine signal paths
 - Solution: add track-and-hold
- Static problems
 - Offsets in folder transistors can cause DNL and INL
 - Interpolation with a factor greater than 2 can introduce DNL and INL due to amplifier nonlinearity
 - Solutions: averaging or calibration